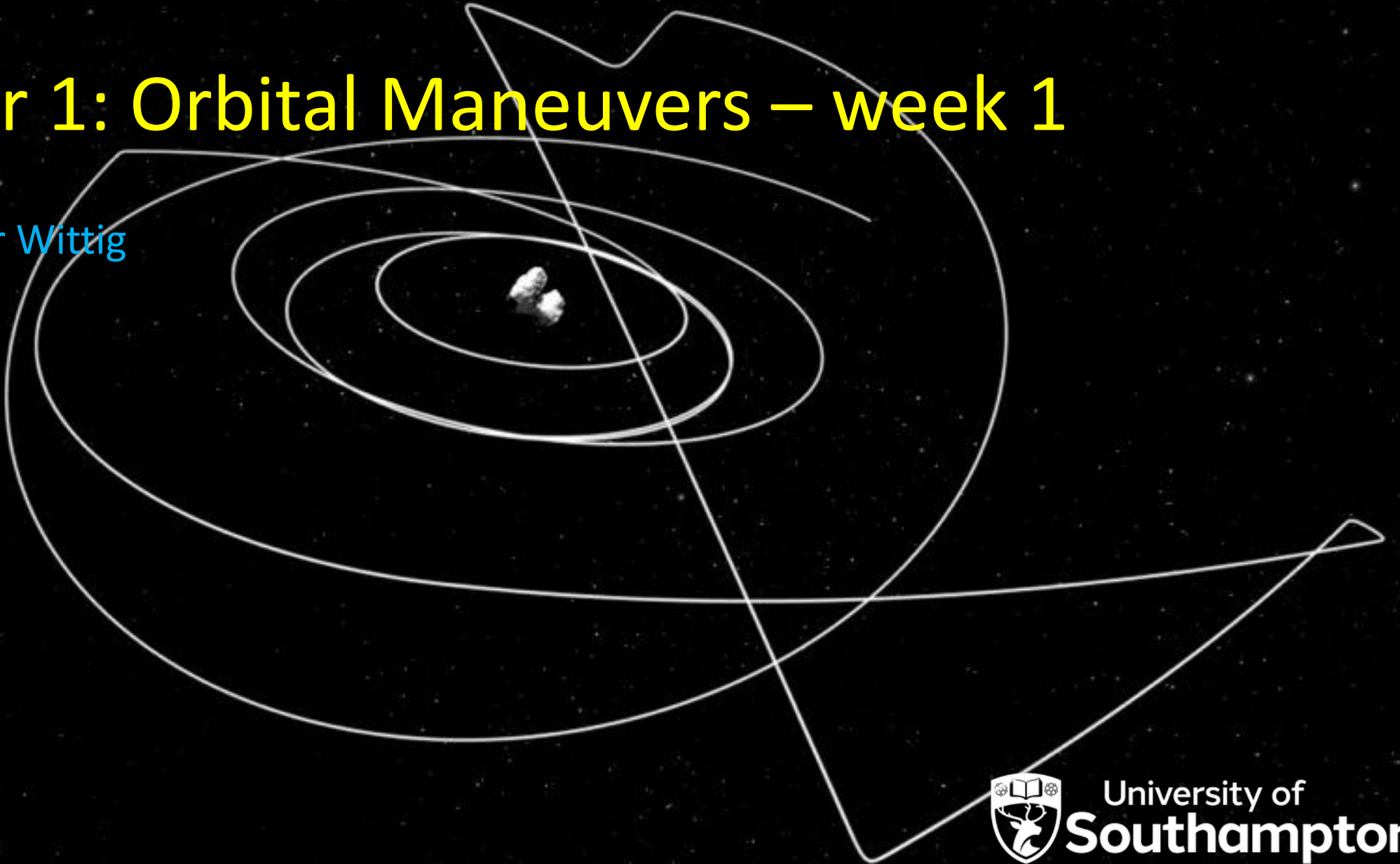


Orbital Mechanics (SESA6076)

Chapter 1: Orbital Maneuvers – week 1

Dr. Alexander Wittig



- **Introduction**
- **Chapter 1: In 20.000 km, turn right. Orbital Manoeuvres**
 - Coplanar Manoeuvres
 - Non-Coplanar Transfers
- **Chapter 2: That's mildly disturbing Orbital Perturbations & Propagation**
 - Non-Uniform Gravity Field
 - Third-Body Perturbation
 - Atmospheric Drag
 - Solar Radiation Pressure
 - Orbit Propagation Methods
- **Chapter 3: Is it a plane? Is it a bird? Orbit Observation & Determination**
 - Observation Techniques
 - Preliminary Orbit Determination
- **Chapter 4: Waltzing Matilda Relative Orbital Motion**
 - Relative Orbital Motion
 - Linearization & State Transition Matrix
- **Chapter 5: To boldly go... Interplanetary Trajectories & Optimization**
 - Lambert's Problem
 - Patched Conics
 - Gravity-assisted fly-by trajectories
- **Chapter 6: Torn between two bodies Restricted Three-Body Problem**
 - N-Body Problem
 - Circular Restricted Three-Body Problem
 - Natural Dynamics around Libration Points
- **Chapter 7: You spin me right round Attitude Dynamics**
 - Attitude representation
 - Attitude Dynamics
- **Chapter 8: In the spin cycle Attitude Motion**
 - Torque-Free Motion
 - Attitude Maneuvers
- **Summary and Revision**

- Design maneuvers to change between co-planar orbits
- Design maneuvers to change orbital plane
- Calculate cost of individual and combined maneuvers
- Identify optimal locations for performing maneuvers depending on type
- Optimize maneuvers for minimal Delta-V

Recall: Impulsive Maneuvers

- Change in impulse assumed delivered instantly
- Change of magnitude and/or direction of velocity
- Cost of mission measured in Δv : change of velocity (vectorial!)

Fuel cost of maneuver with given Δv (Tsiolkovski):

$$m_f = m_i - \Delta m = m_i \exp \left(-\frac{\Delta v}{I_{sp} g_0} \right)$$

m_i , m_f : initial and final spacecraft mass

I_{sp} : specific impulse of engine

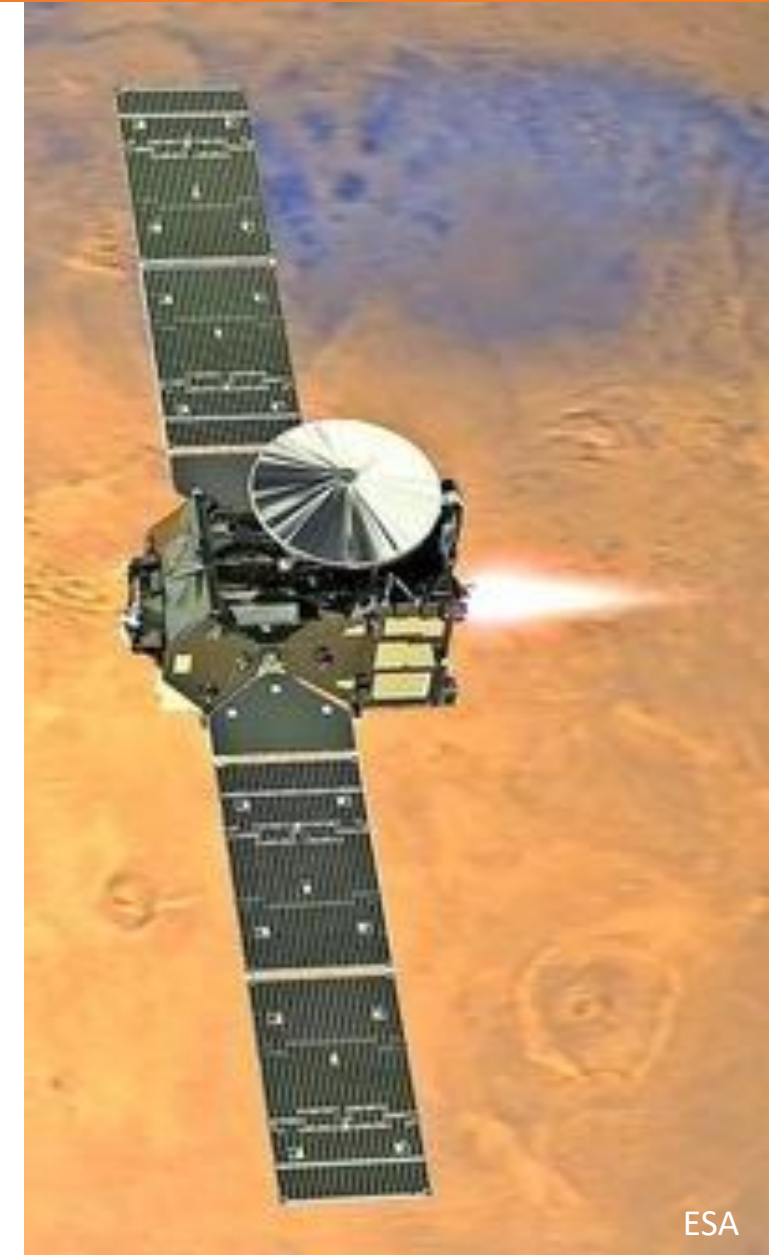
g_0 : standard Earth gravity at sea level (9.80665 m/s²)

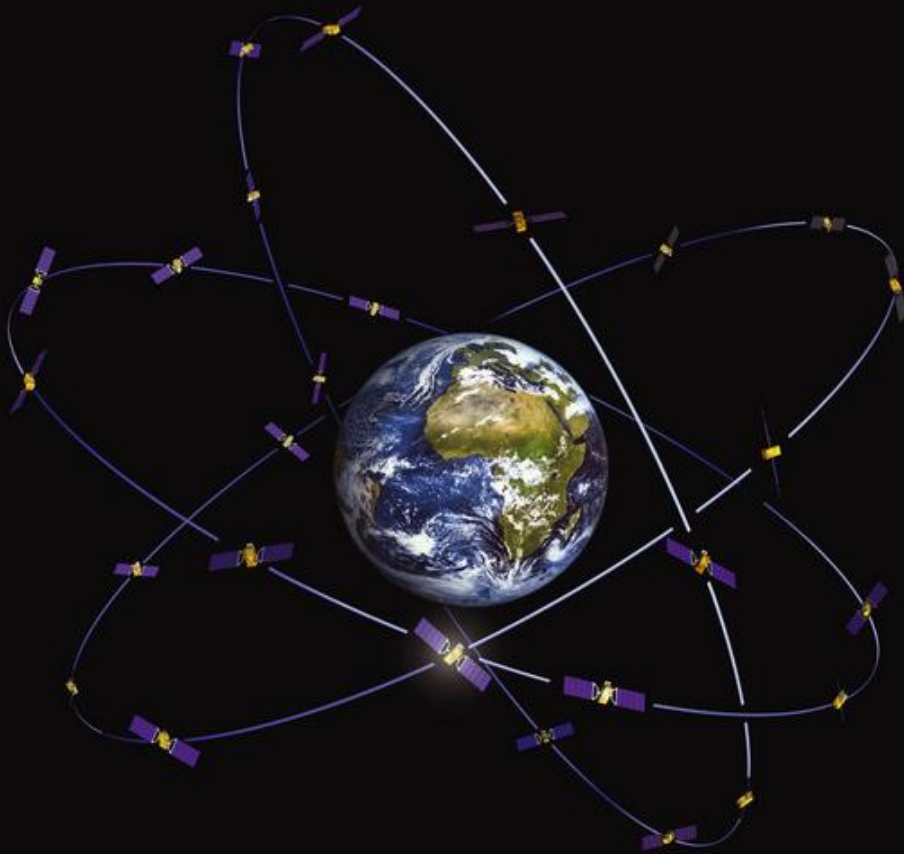


Note

Fuel cost is additive in Δv ,
i.e. Δv of multiple burns can
be summed up to Δv_{tot} .

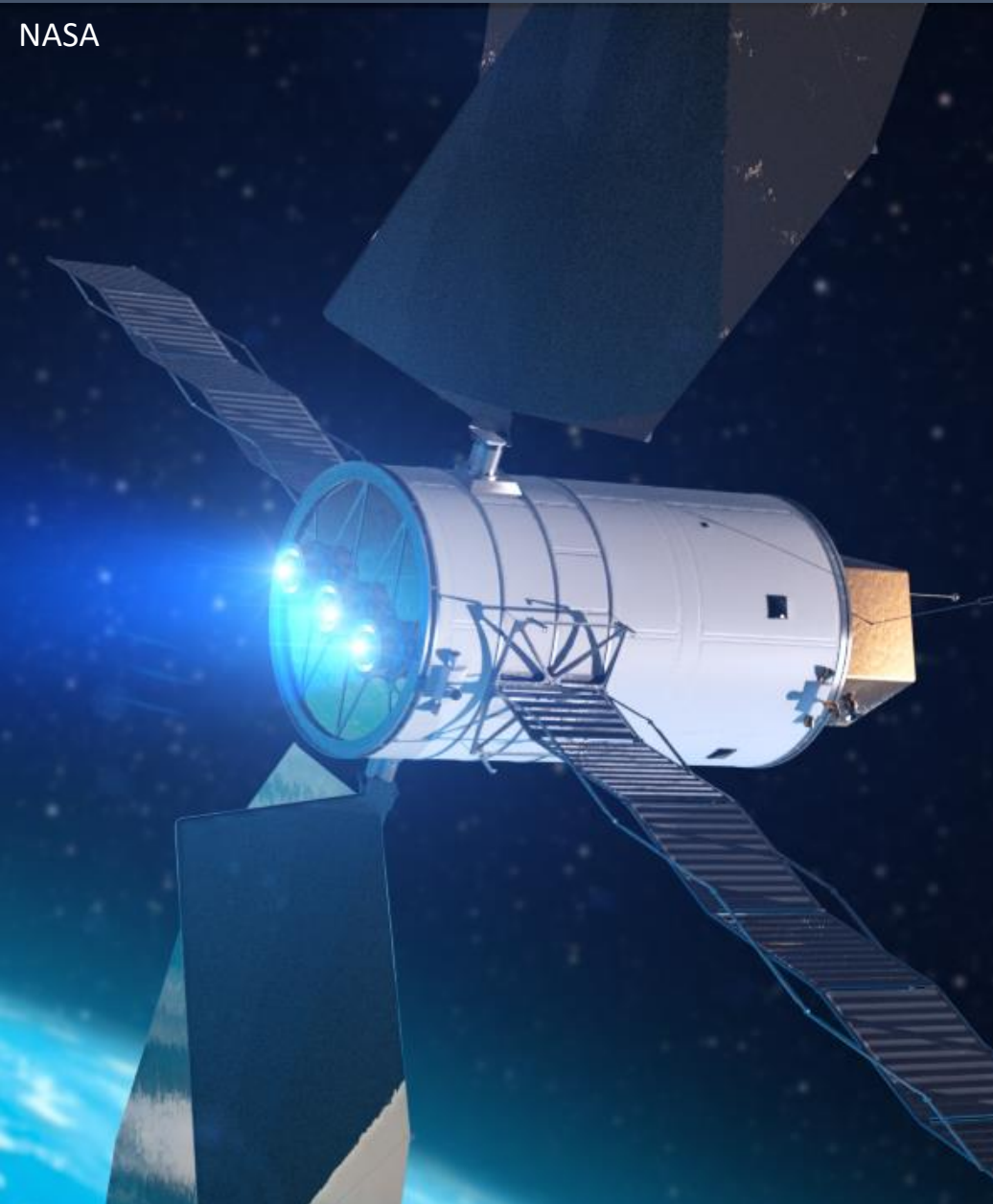
$$\begin{aligned} m_{f,1} &= m_{i,1} \exp \left(-\frac{\Delta v_1}{I_{sp} g_0} \right) = m_{i,2} \\ m_{f,2} &= m_{i,2} \exp \left(-\frac{\Delta v_2}{I_{sp} g_0} \right) \\ &= m_{i,1} \exp \left(-\frac{\Delta v_1}{I_{sp} g_0} \right) \exp \left(-\frac{\Delta v_2}{I_{sp} g_0} \right) \\ &= m_{i,1} \exp \left(-\frac{\Delta v_1 + \Delta v_2}{I_{sp} g_0} \right) \end{aligned}$$





ESA Galileo

- Maneuver made of building blocks
 - Change orbit shape
 - Change orbit orientation (plane)
 - Change of orbit phasing (anomaly)
- Multiple maneuvers throughout mission
- Orbit transfer / maintenance / disposal
- Goal of mission design: minimize Δv within mission constraints
 - Transfer time
 - Forbidden regions (e.g. radiation belts)
 - Pointing requirements (panels, instruments)



- Coplanar transfers
 - (Bi-elliptic) Hohmann transfer
 - Phasing maneuvers
 - Non-Hohmann transfers
 - Apse line rotation
- Plane change maneuvers
 - Inclination change
 - RAAN change

Coplanar Maneuvers

Hohmann transfer

- Change from one orbit shape to another
- Usually during launch & commissioning phase: move satellite in place
- Deep space missions: sequence of transfers to target body
- In practice combined with inclination change (if required)
- Initial delta-V can be supplied by launcher (staging!)
- Examples:
 - Parking orbit departure
 - GTO insertion



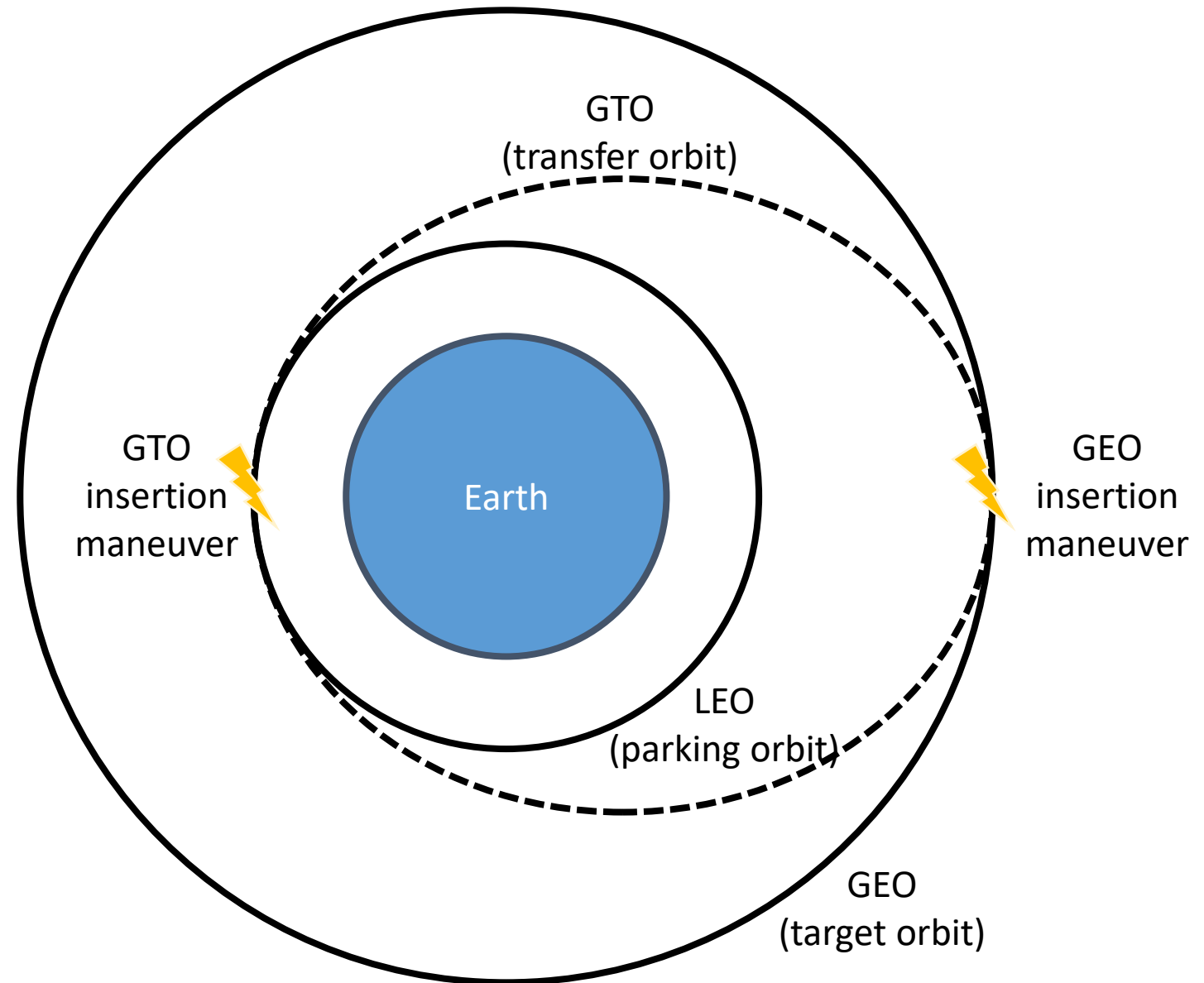
- Transfer between distinct orbits requires two burns

- Consider energy:

$$\epsilon = -\frac{\mu}{2a}$$

- Increase a = increase energy
- Both burns increase velocity

Hohmann transfer



Classic Hohmann Transfer

Transfer between circular orbits with r_1 and r_2

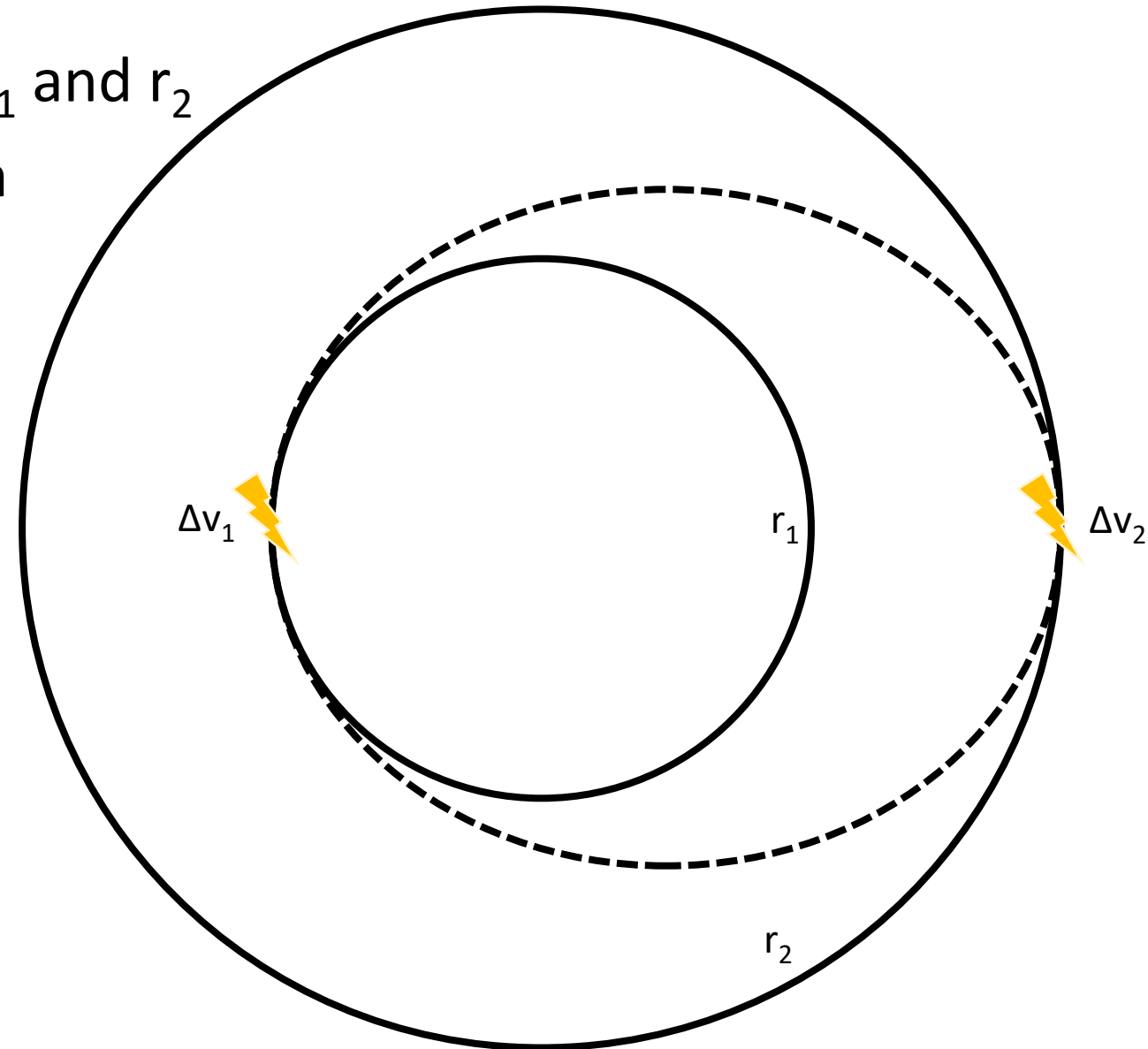
- Δv determined from energy equation
- Or using elliptic angular momentum:

$$r_p = \frac{h^2 / \mu}{1 + e} \quad e = \frac{r_a - r_p}{r_a + r_p}$$

$\Rightarrow$

$$h^2 = \mu r_p \left(1 + \frac{r_a - r_p}{r_a + r_p} \right)$$

$$h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}}$$



Transfer between circular orbits with r_1 and r_2

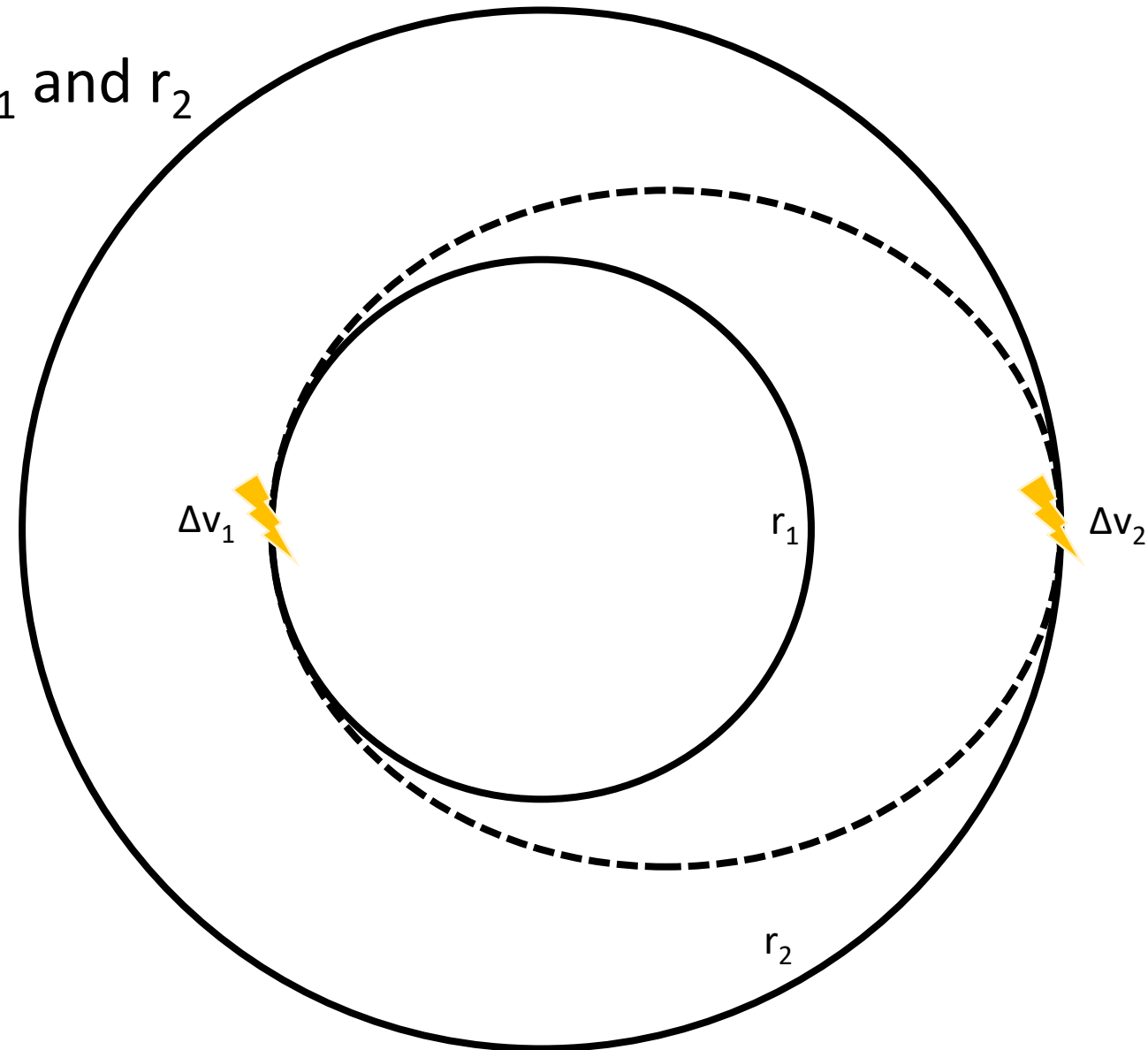
$$h = \sqrt{2\mu \frac{r_a r_p}{r_a + r_p}}$$

$$v_a = h/r_a$$

$$v_p = h/r_p$$

Special case of circular orbits:

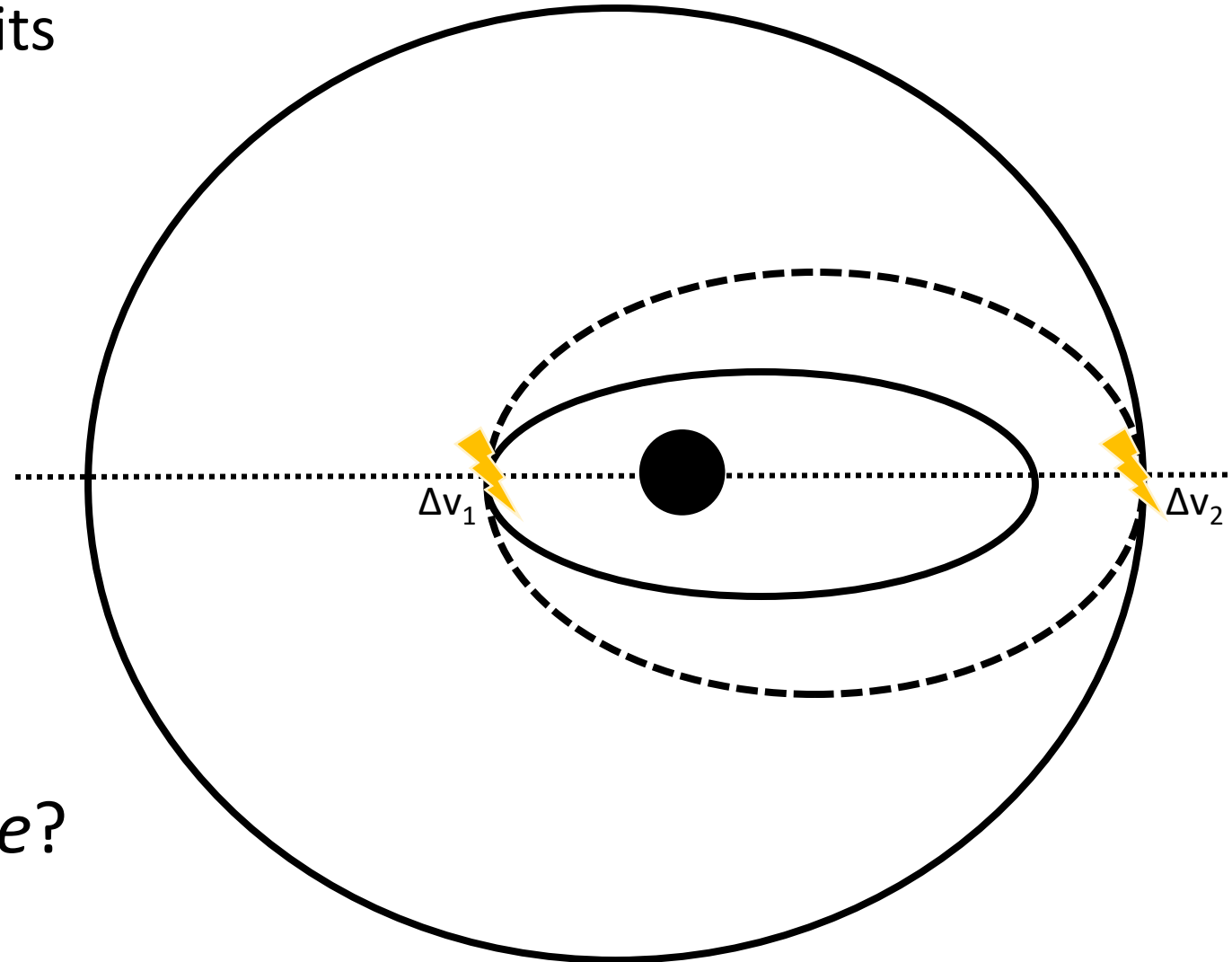
$$h = \sqrt{\mu r}$$



- Nothing restricts us to circular orbits
- **Coaxial** ellipses are fine as well
 - To maintain tangential transfer orbit
- Same principle
- Same equations apply

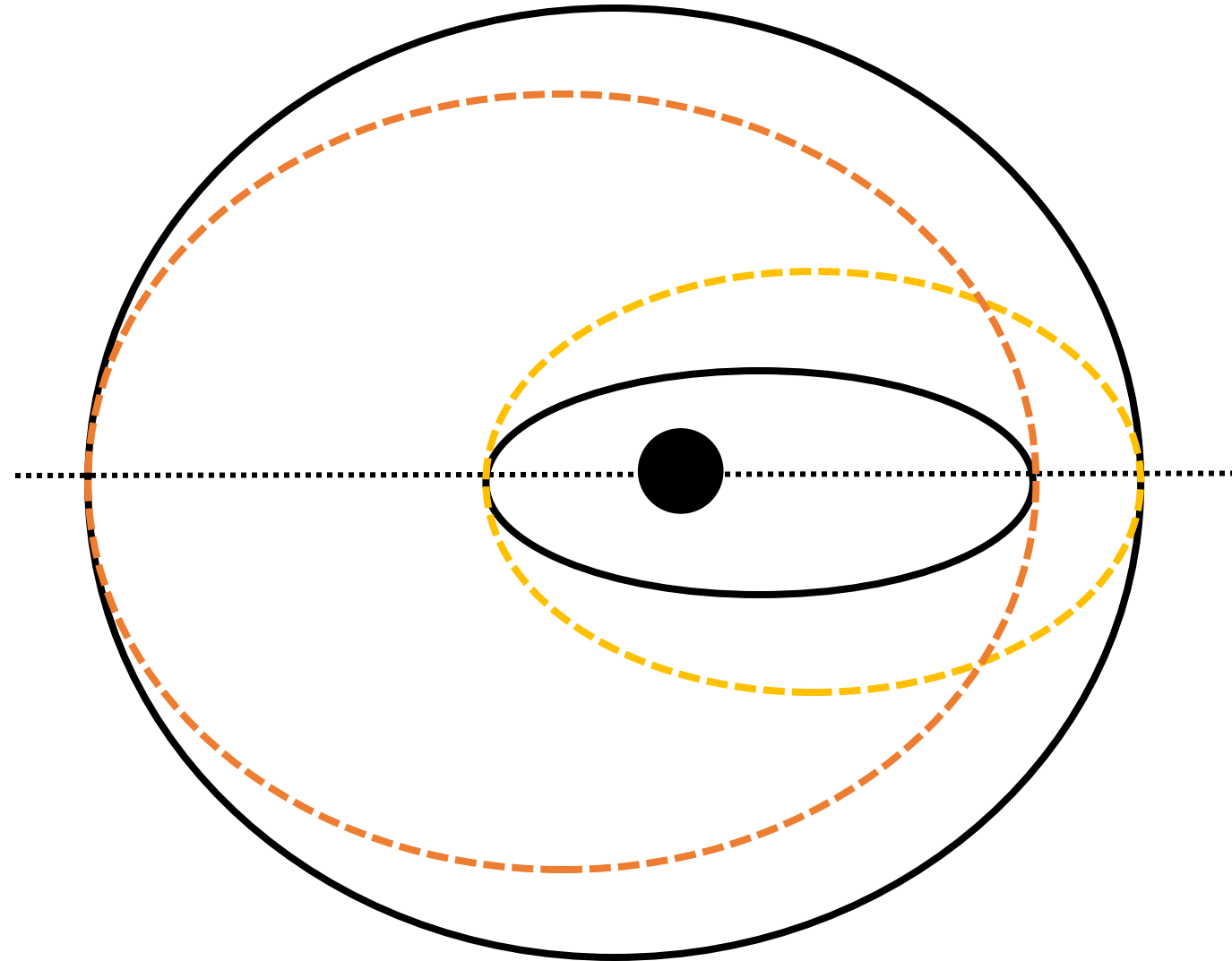
How to find h for orbit given a , e ?

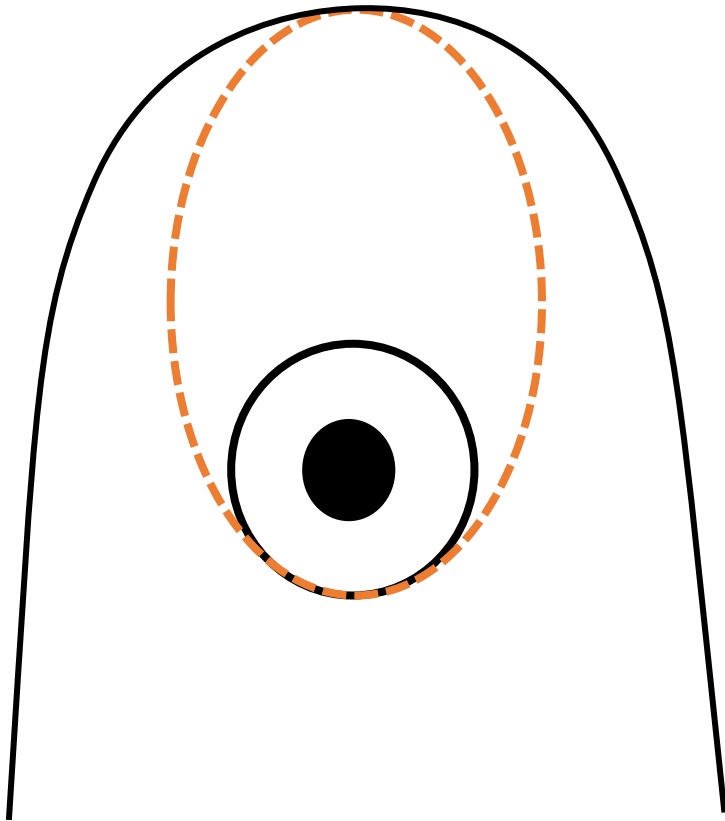
$$h^2 / \mu = a(1 - e^2)$$



- Two distinct elliptic transfers
 - inner Apogee
 - inner Perigee
- Not obvious which is better

Need to test both transfers



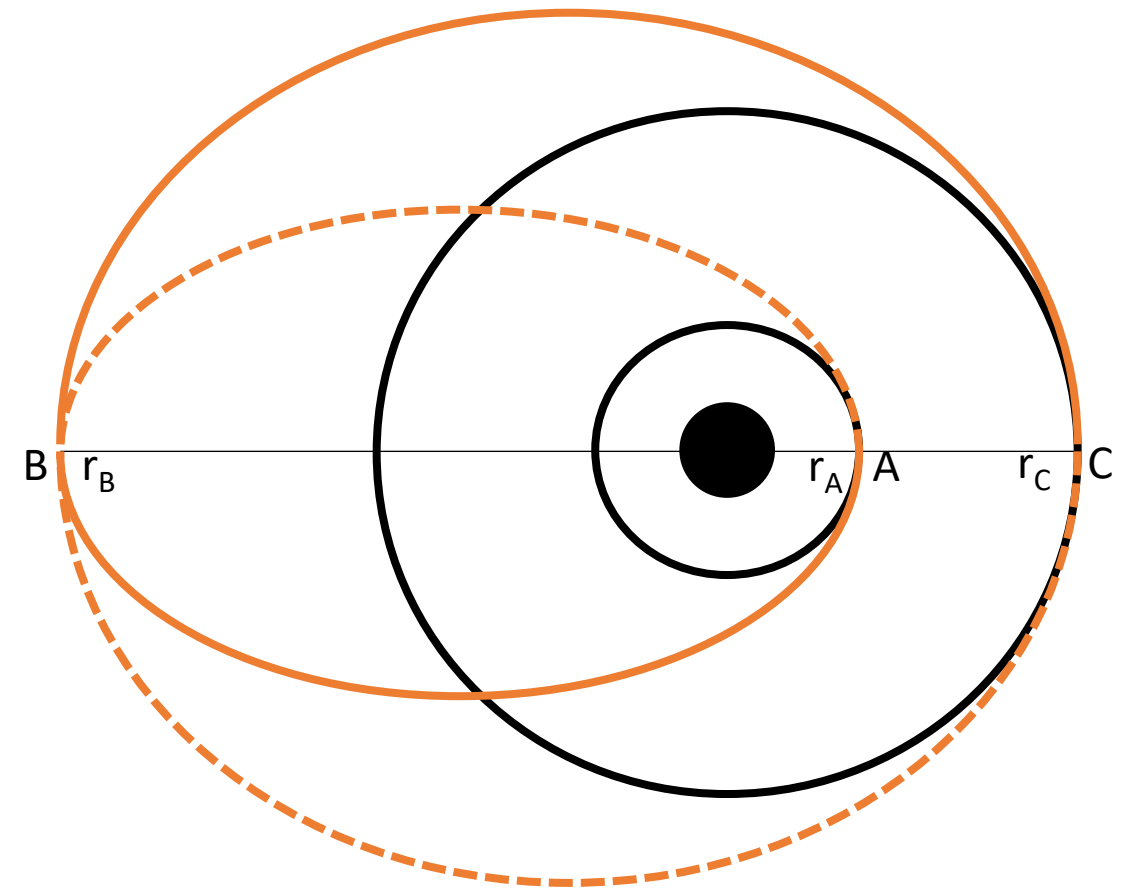


- Transfer ellipse tangential to both orbits
- Transfer always takes $\frac{1}{2}$ orbit of transfer ellipse
- Optimal **2-burn** transfer between two **coaxial** ellipses (mind the constraints!)
- Cost of backward transfer same as forward transfer (symmetry!)
- Can also be extended to transfers from/to periapsis of non-elliptic orbits

Coplanar Maneuvers

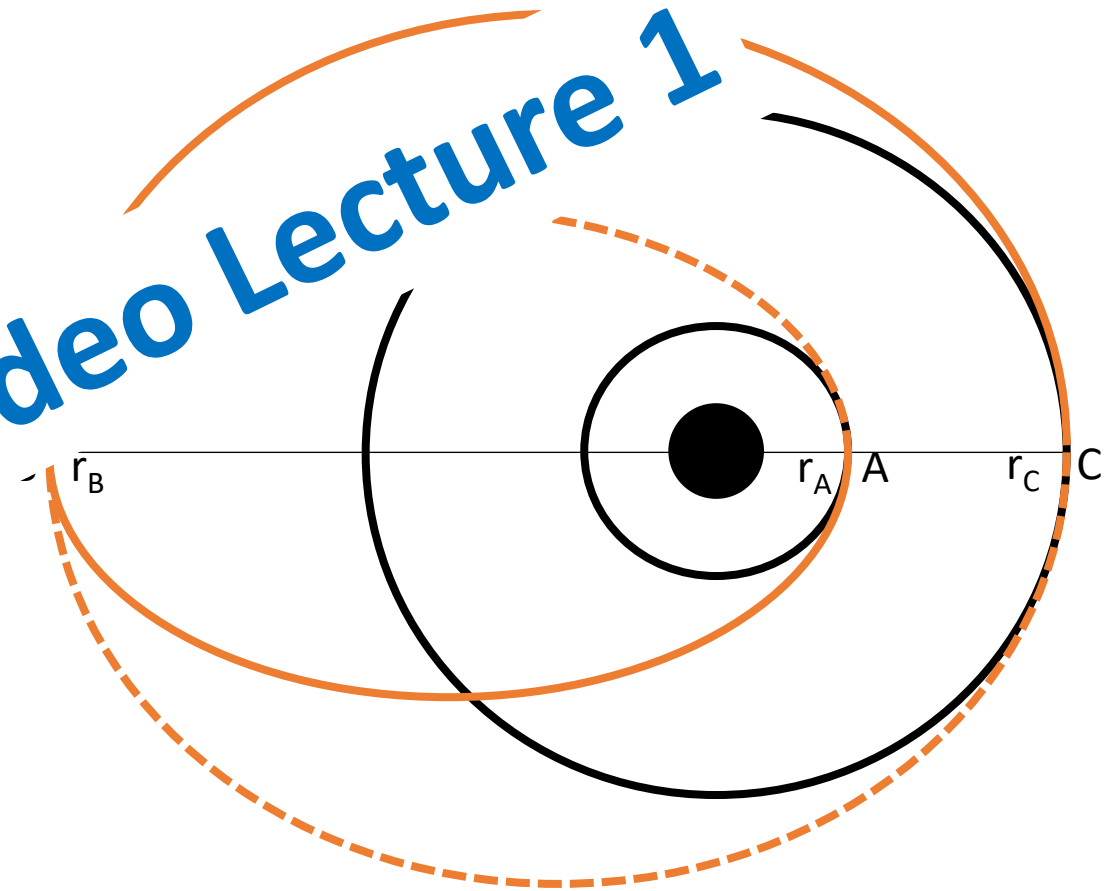
Bielliptic Hohmann transfer

- Two **circular** orbits
- Two Hohmann transfers via exterior point
- As r_B goes to infinity, Δv required at B goes to zero
- Can be more fuel-efficient depending on r_B

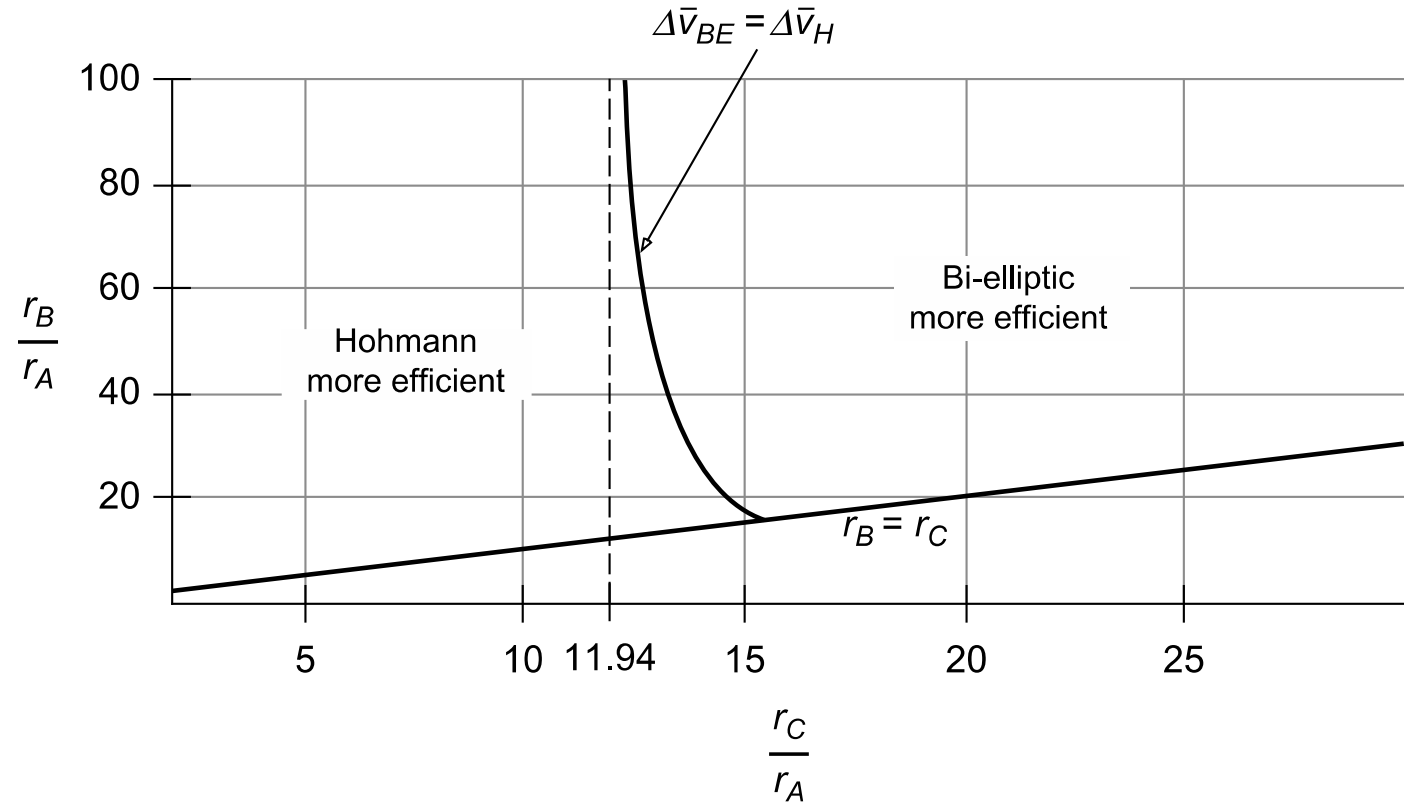
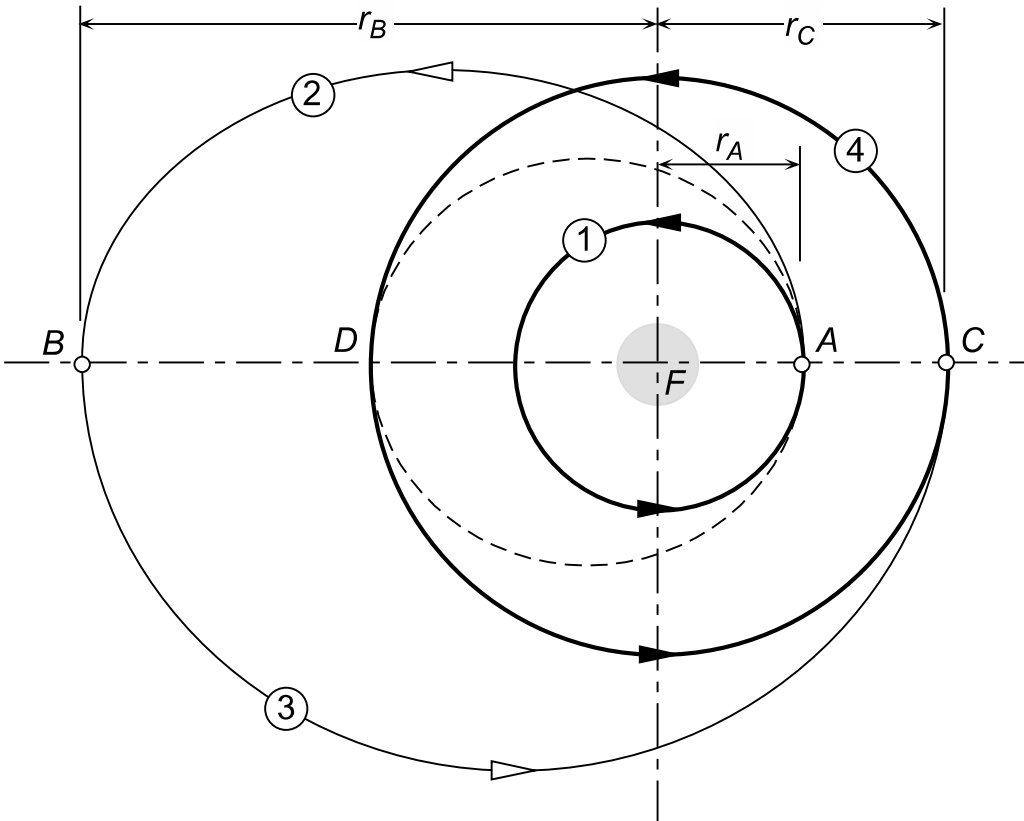


- Two **circular** orbits
- Two Hohmann transfers via exterior point
- As r_B goes to infinity, Δv required at B goes to zero
- Can be more efficient depending on

see Chapter 1 Video Lecture 1



Bi-elliptic vs. Classical Hohmann

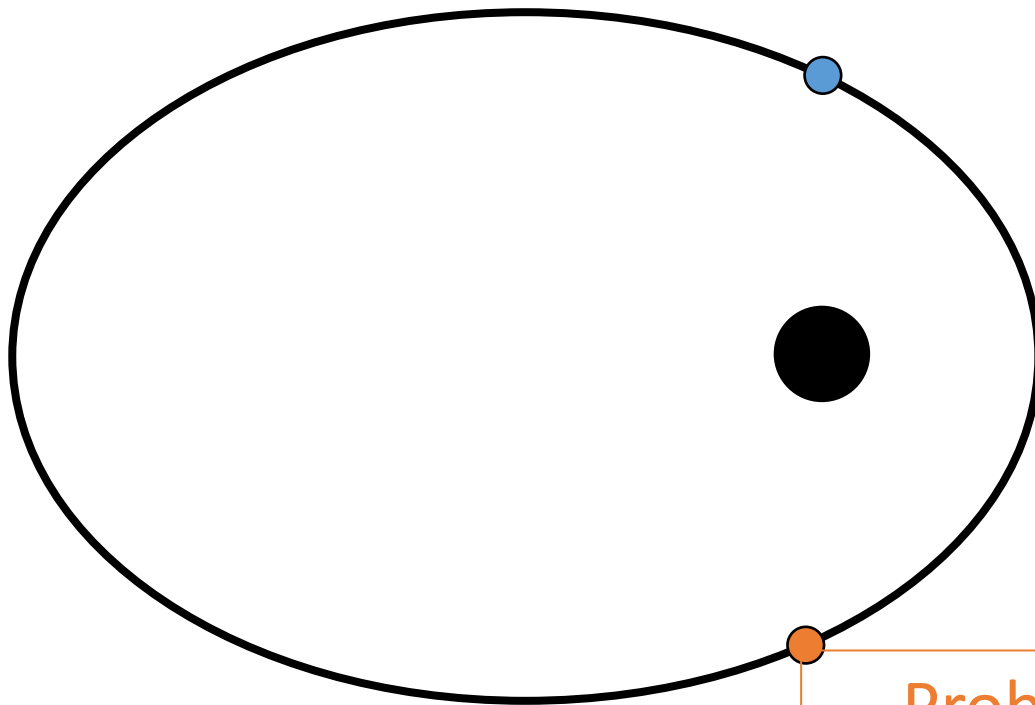


Curtis

Coplanar Maneuvers

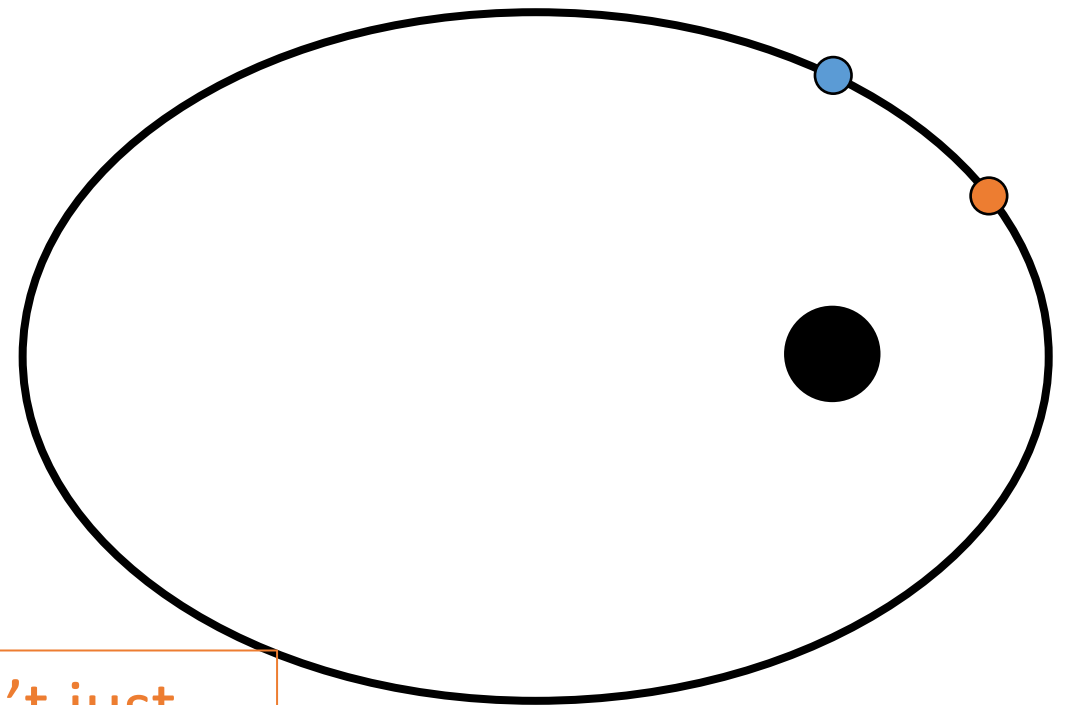
Phasing maneuvers

Goal: Change **phase** (offset to true anomaly) in a given orbit.



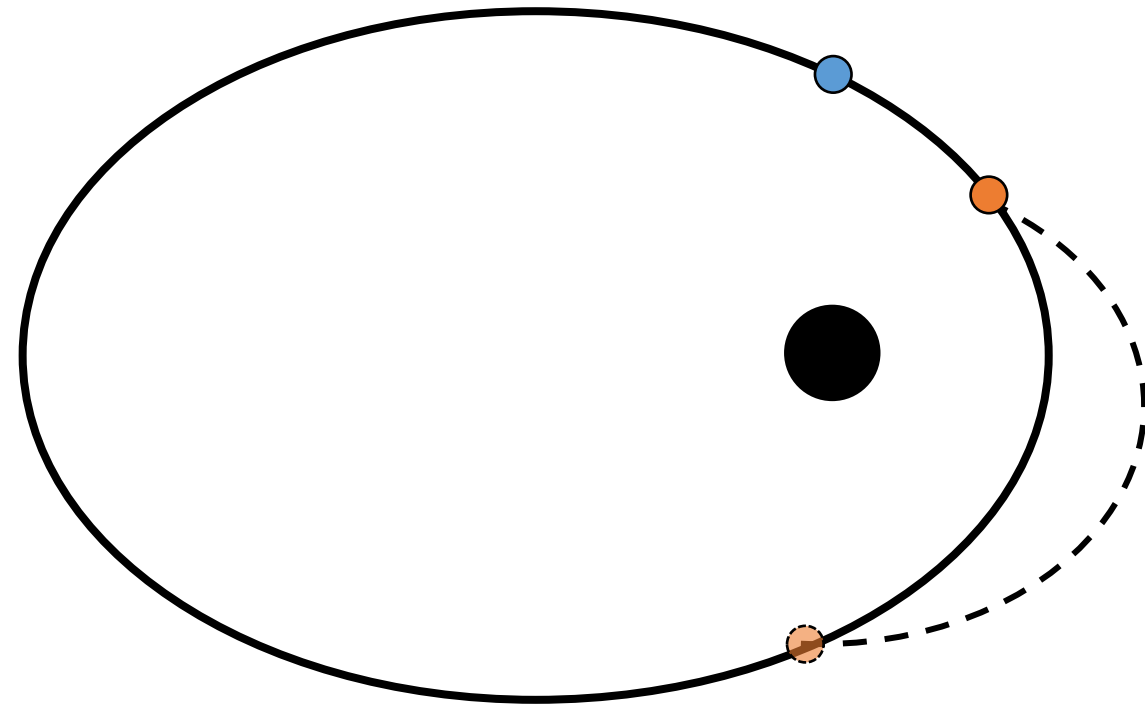
Before Maneuver

Problem: can't just
speed up along orbit

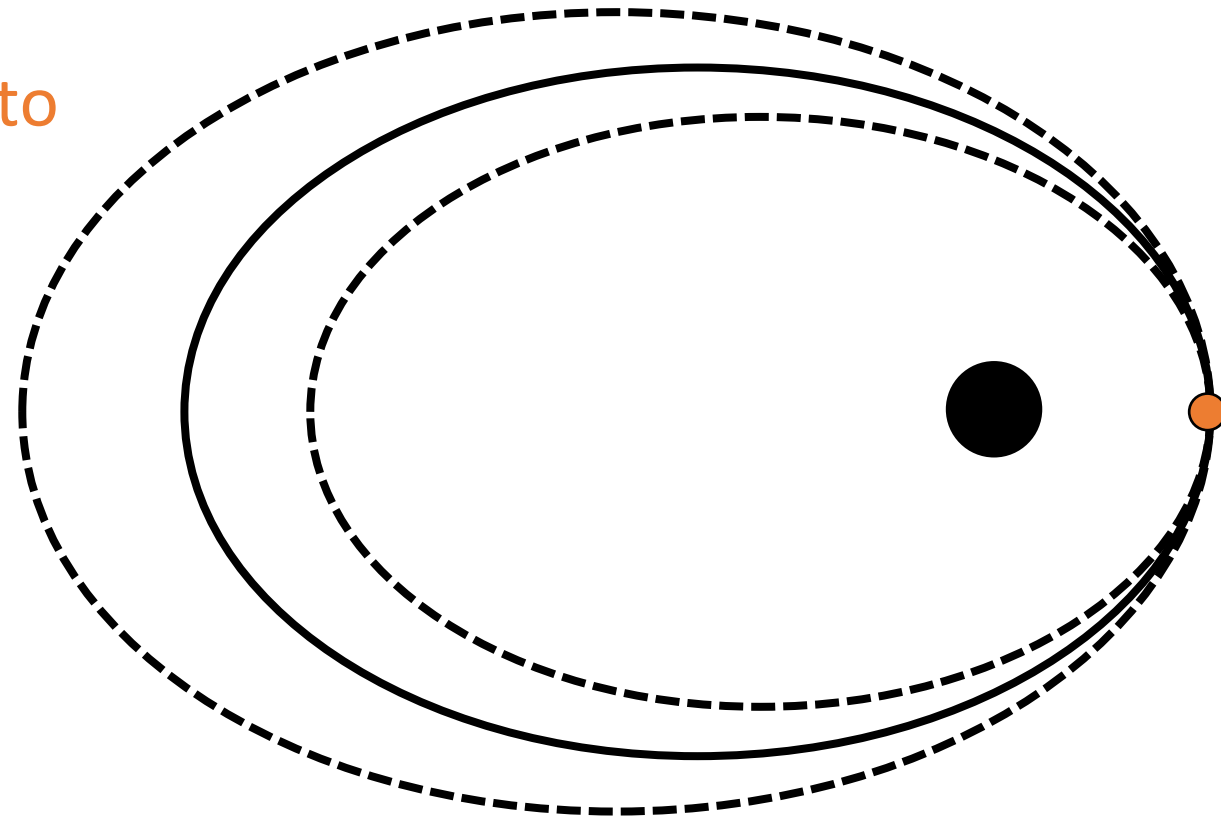


After Maneuver

- Could try to find orbit to get chaser from old position to new position in time for target to do one turn (Lambert solver!)



- ~~■ Could try to find orbit to get chaser from old position to new position in time for target to do one turn (Lambert solver!)~~
- Easier, explicit solution:
Two-impulse Hohmann transfer from/to same orbit
- Get back to same point in orbit faster / slower to change phasing



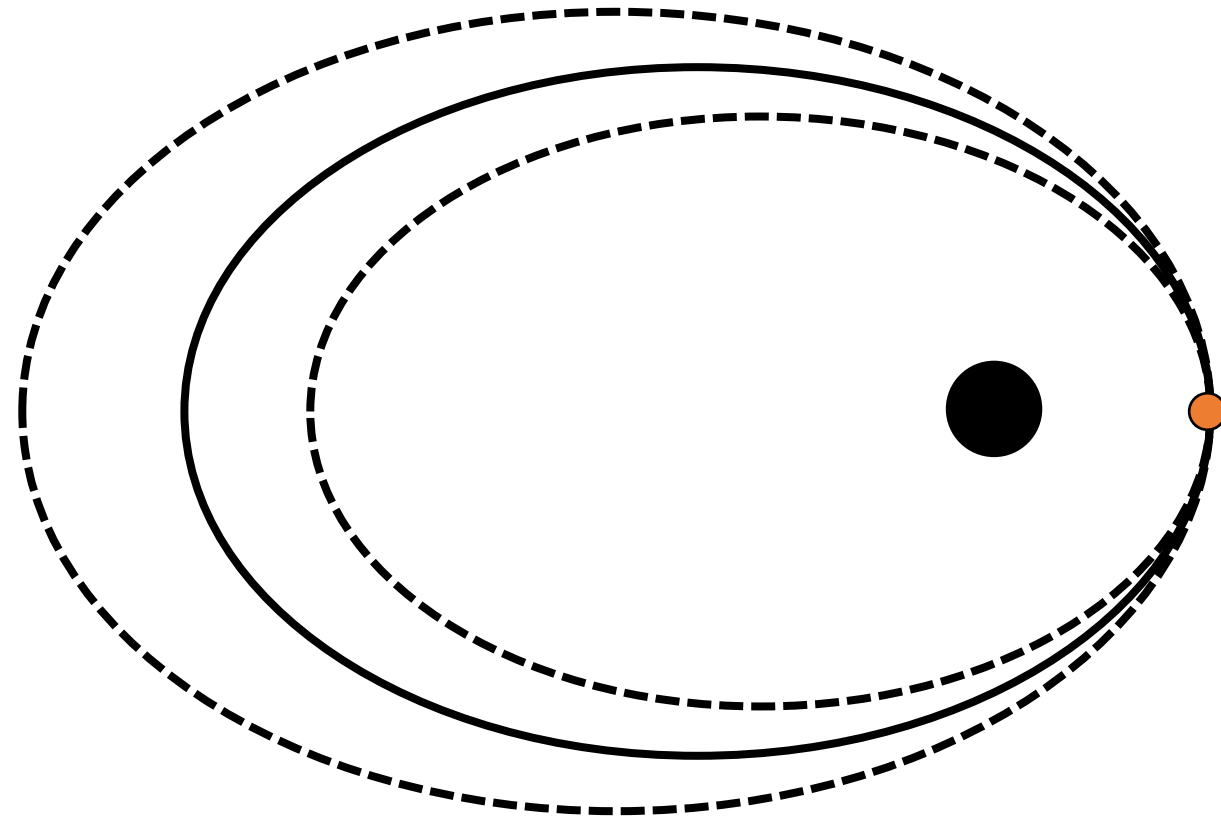
1. Determine period T of phasing orbit
2. Compute a :

$$a = \left(\frac{T \sqrt{\mu}}{2\pi} \right)^{\frac{2}{3}}$$

3. Compute apogee/perigee radius:

$$2a = r_a + r_p$$

4. Set up Hohmann transfer orbit
(both DV same magnitude but opposite sign)



- retro-fire (slow down) to lower semi-major axis (speed up)
- forward-fire (speed up) to raise semi-major axis (slow down)
- Uses:
 - chase spacecraft
 - rendezvous
 - move Earth-relative location (e.g. in GEO)
- reduce cost by allowing phasing over multiple revolutions

- retro-fire (slow down) to lower semi-major axis (speed up)
- forward-fire (speed up) to raise semi-major axis (slow down)
- Uses:
 - chase spacecraft
 - rendezvous
 - move Earth-relative location

- reduce cost
 - phasing over multiple revolutions

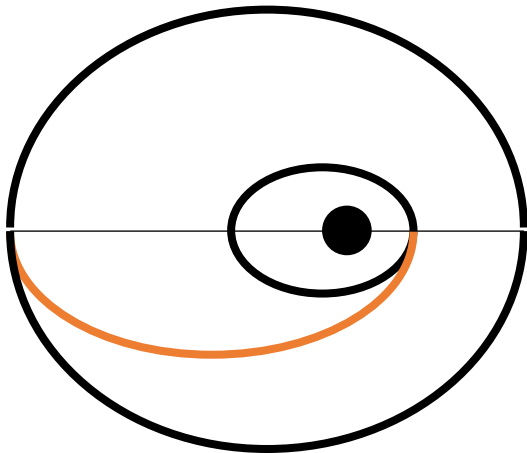
see Chapter 1 Video Lecture 2

Coplanar Maneuvers

Non-Hohmann transfers

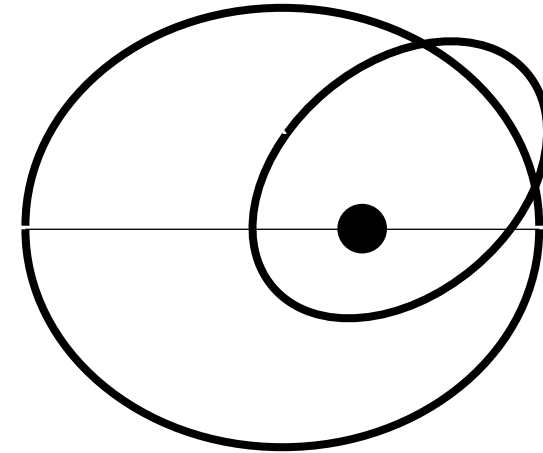
Hohmann

- efficient
- apsis to apsis (to apsis)
- co-axial
- Δv aligned with orbital velocity



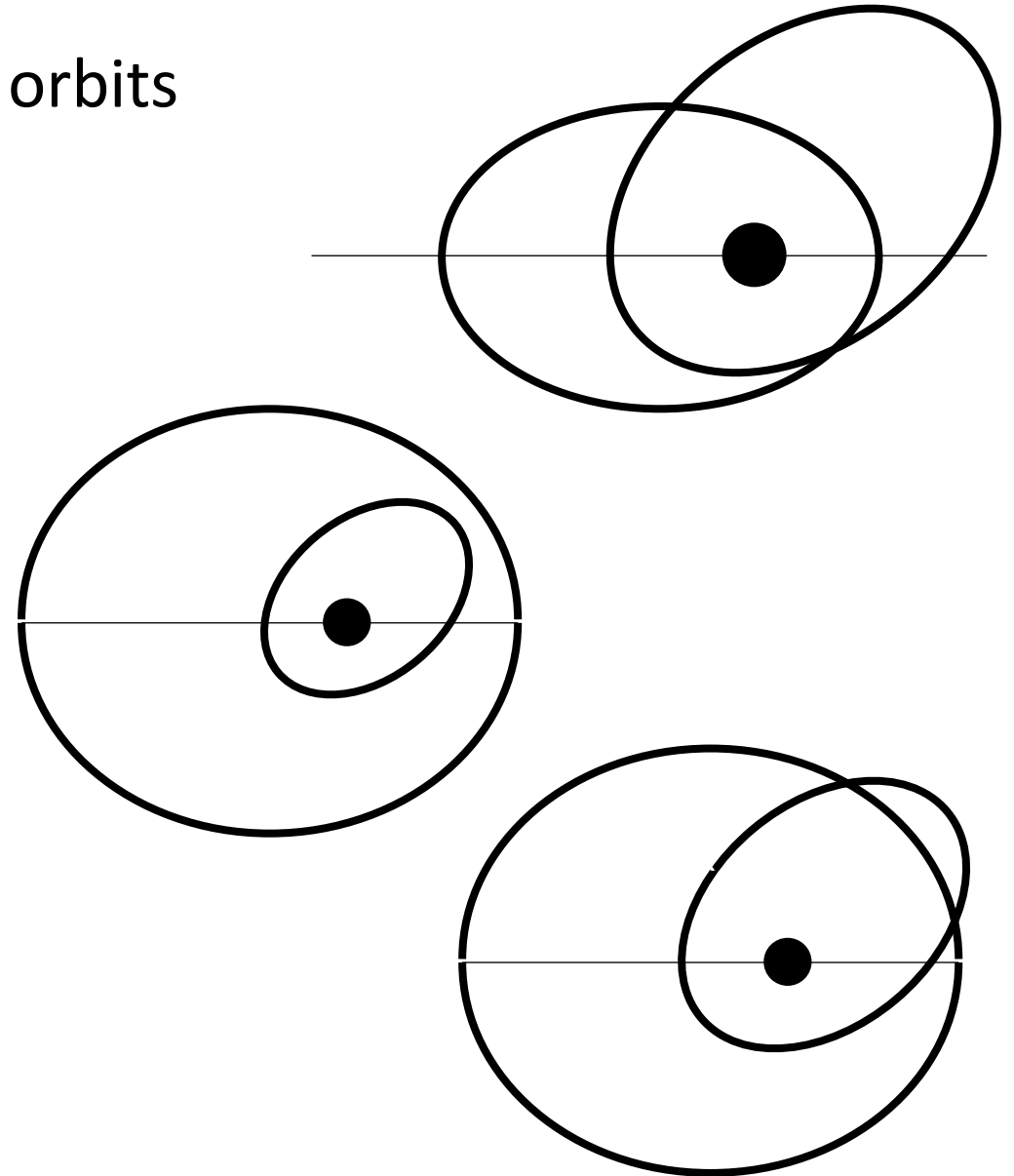
Non-Hohmann

- less efficient
- between any points in orbit
- arbitrary orientation
- Δv vector rotates: higher cost!



Not all transfers can be achieved with Hohmann!

- Arbitrary maneuver for switching between orbits
- Position fixed in impulsive maneuver → only intersecting orbits
- Non-intersecting orbits: two maneuvers & transfer orbit
- Examples:
 - Apse line rotation
 - Plane change
 - Combined maneuvers

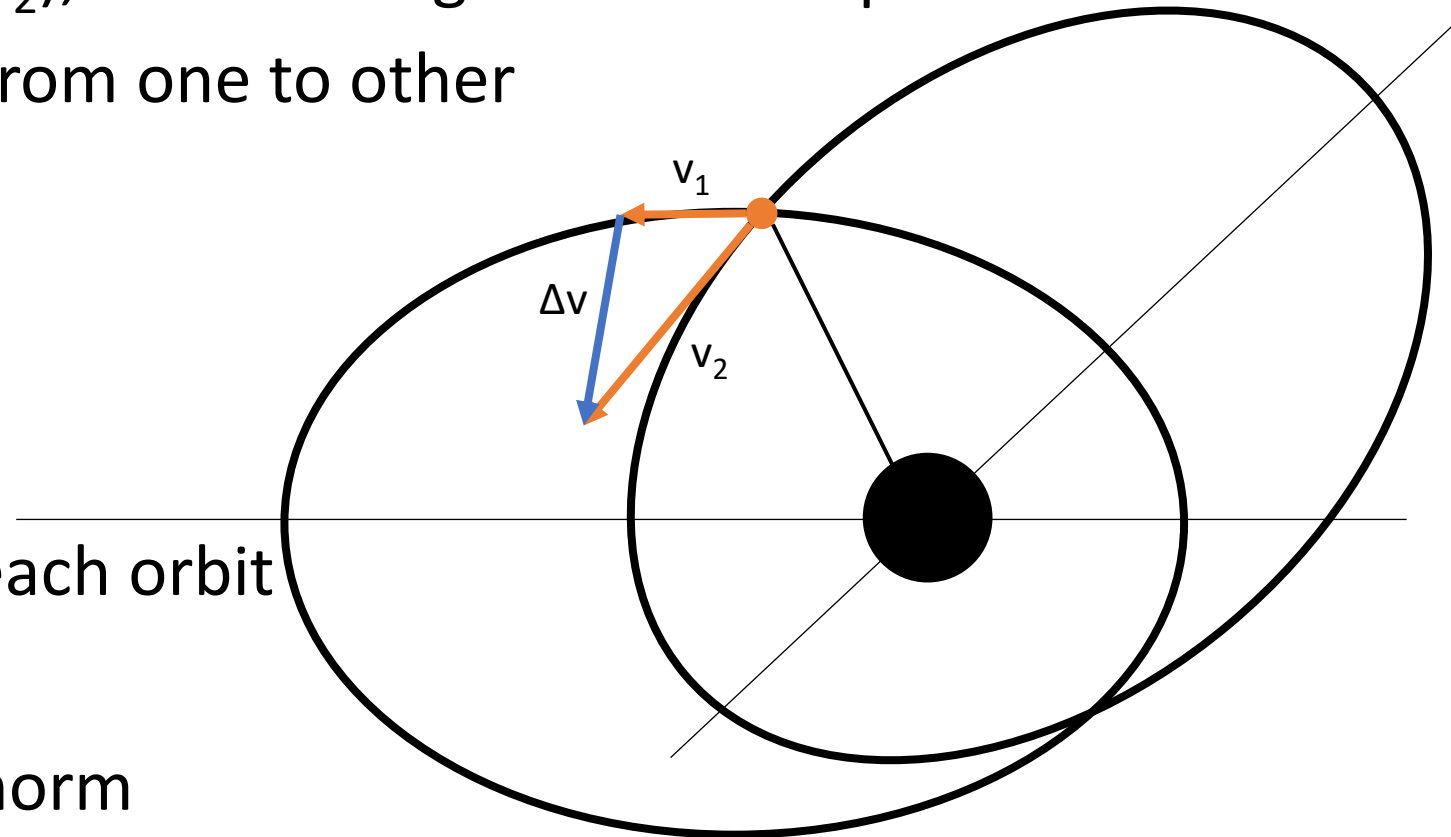


Single impulse apse line rotation (and optional shape change)

Known: orbits 1 & 2 (e_1, h_1, e_2, h_2), rotation angle between apse lines

Wanted: required Δv to switch from one to other

1. Find intersection point
2. Calculate velocity vector in each orbit
3. Calculate difference vector norm



Single impulse apse line rotation (and optional shape change)

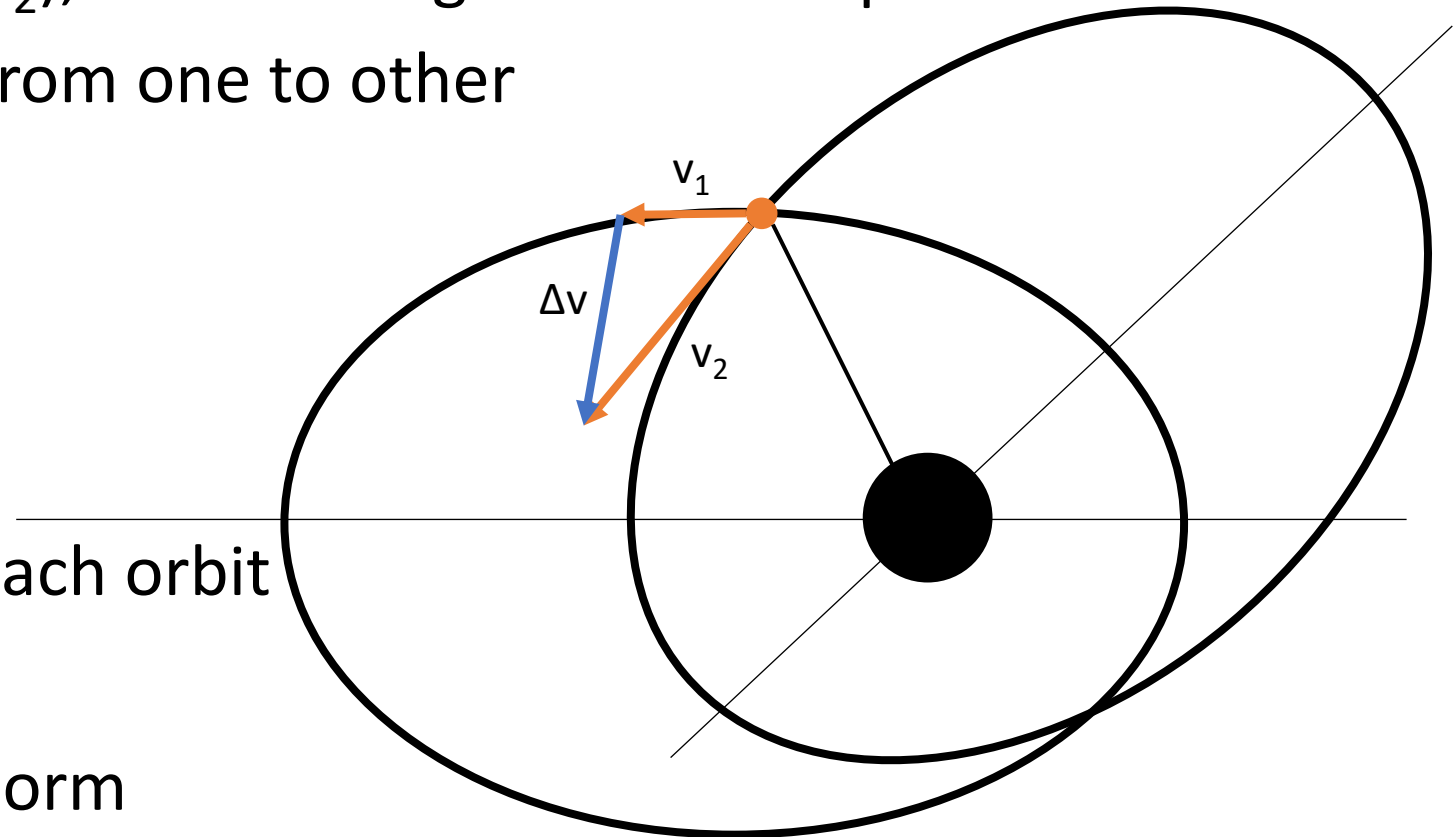
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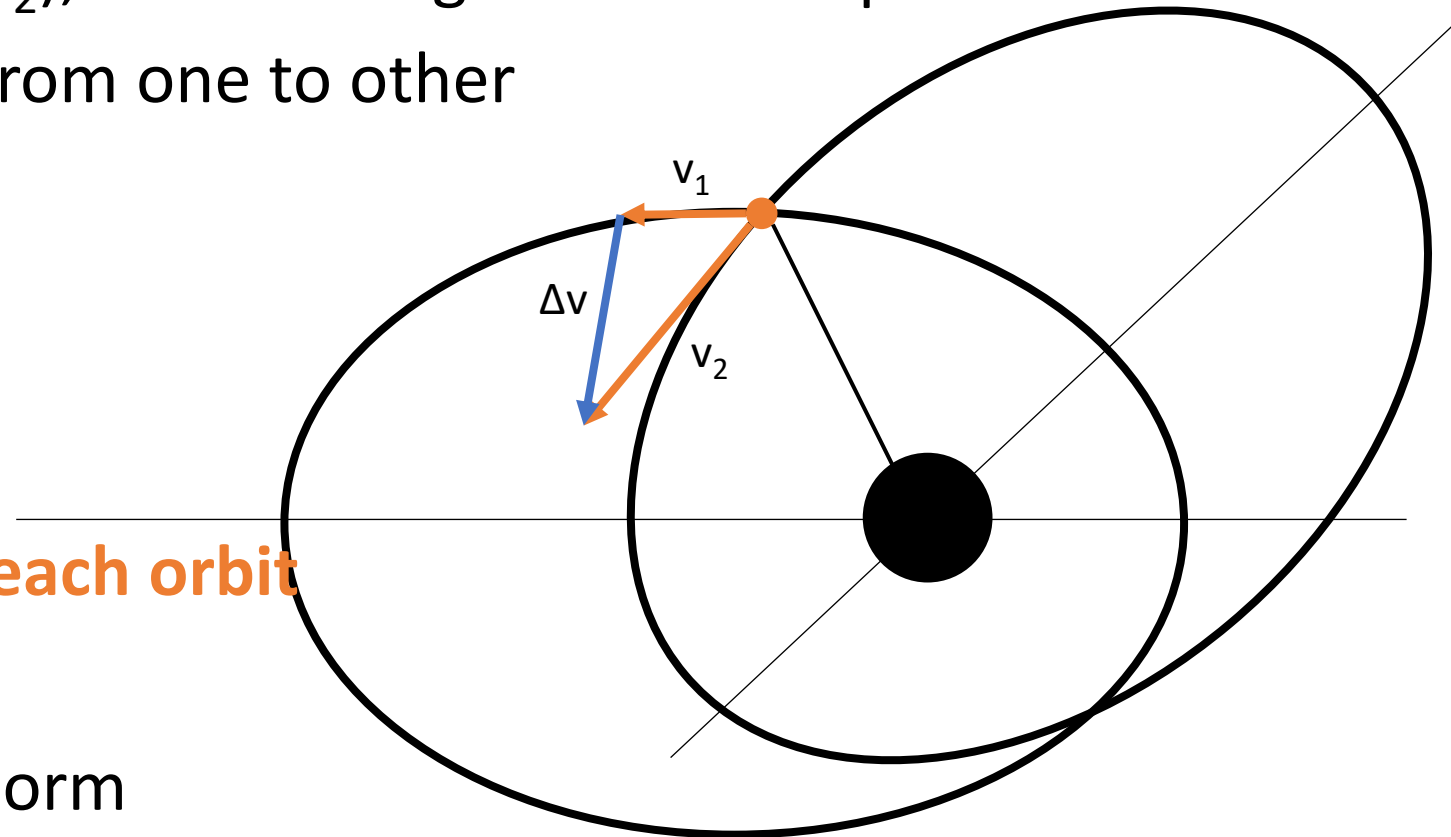


Single impulse apse line rotation (and optional shape change)

Known: orbits 1 & 2 (e_1, h_1, e_2, h_2), rotation angle between apse lines

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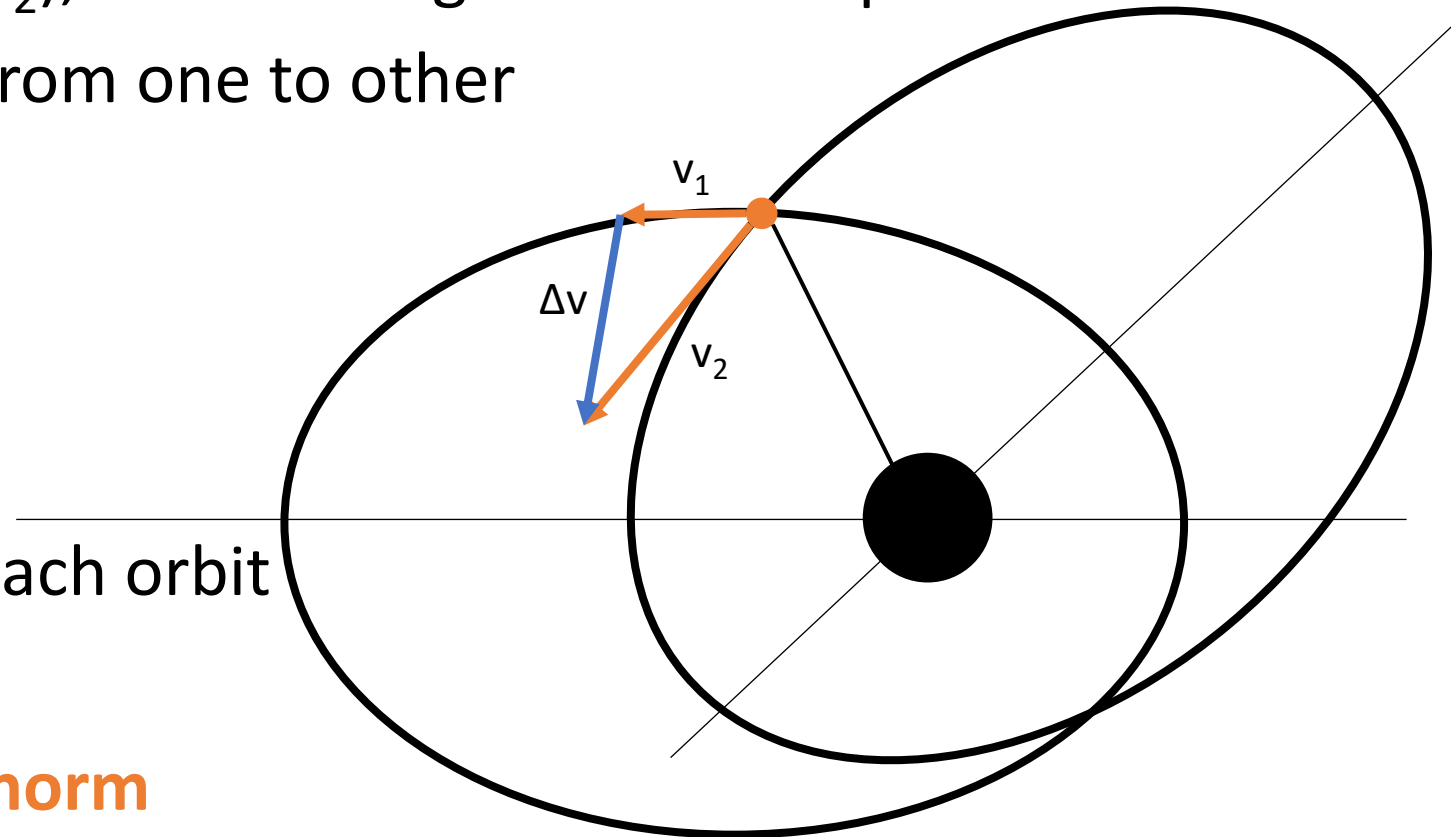


Single impulse apse line rotation (and optional shape change)

Known: orbits 1 & 2 (e_1, h_1, e_2, h_2), rotation angle between apse lines

Wanted: required Δv to switch from one to other

1. Find intersection point
2. Calculate velocity vector in each orbit
3. Calculate difference vector norm



Apse line rotation (single impulse)

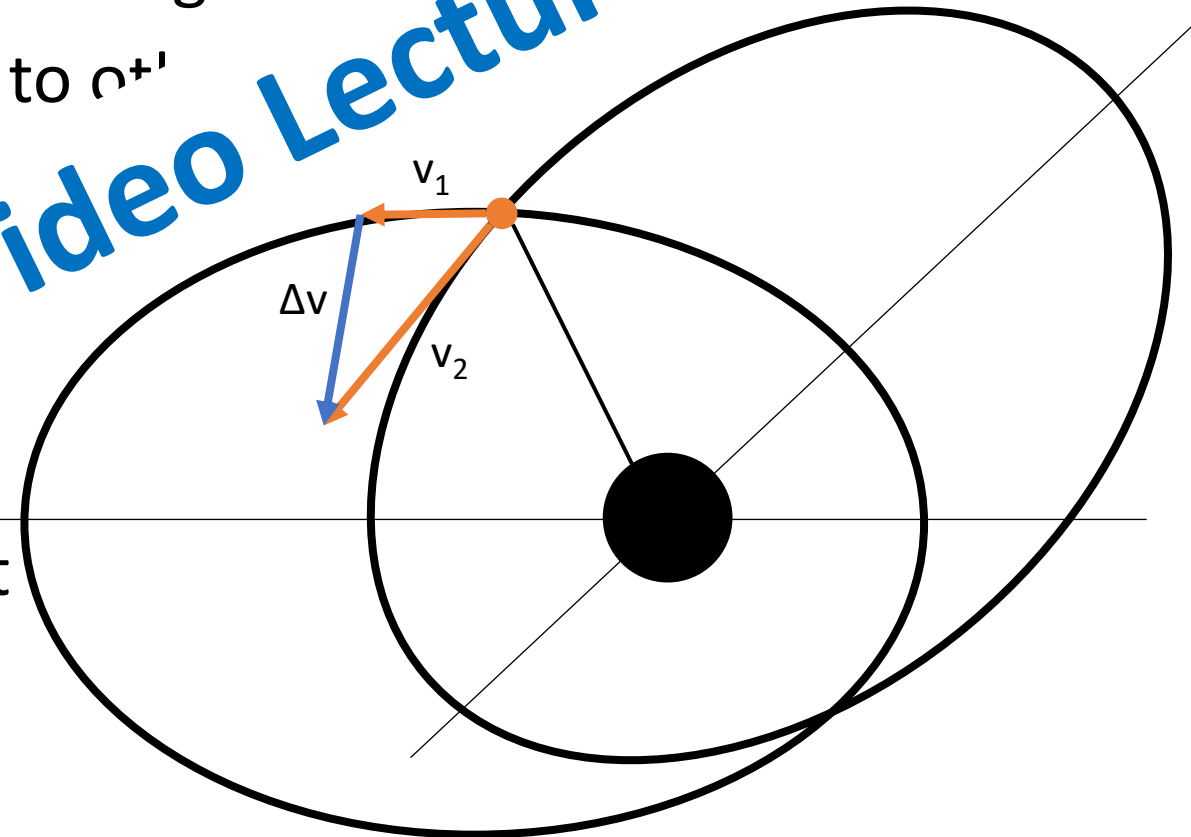
Single impulse apse line rotation (and optional shape change)

Known: orbits 1 & 2 (e_1, h_1, e_2, h_2), rotation angle β

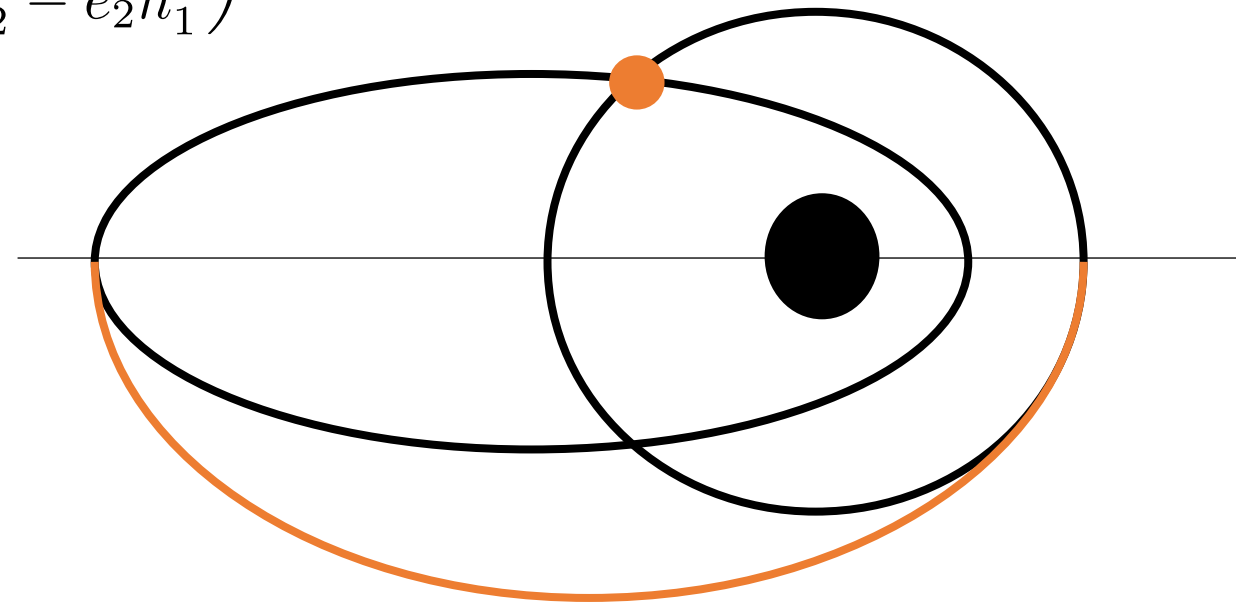
Wanted: required Δv to switch from one to other

1. Find intersection point
2. Calculate r and v in each orbit
3. Calculate difference vector norm

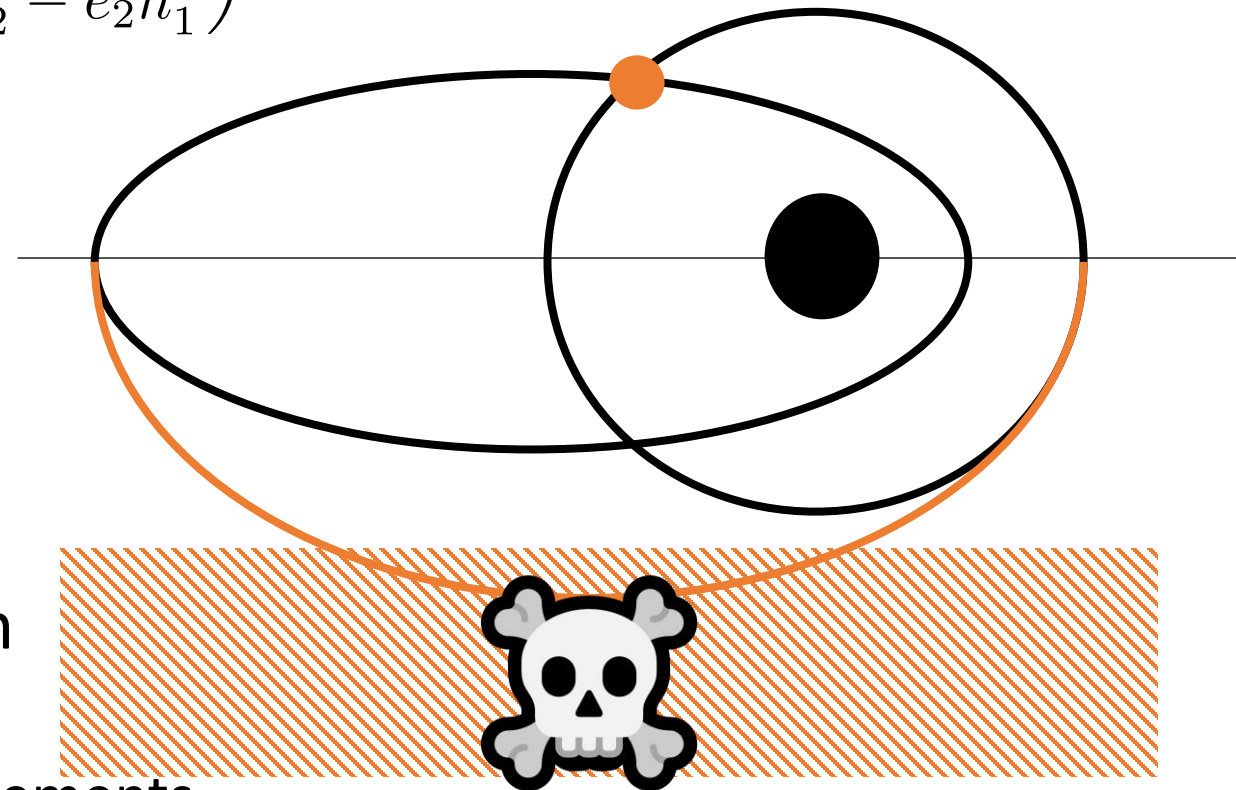
see Chapter 1 Video Lecture 3



- Special case: no apse line rotation
 - Simpler expression for $\theta_1 = \arccos \left(\frac{h_1^2 - h_2^2}{e_1 h_2^2 - e_2 h_1^2} \right)$
 - Rest remains unchanged
- Coaxial transfer with single impulse
 - Less efficient than Hohmann
 - Requires only one burn

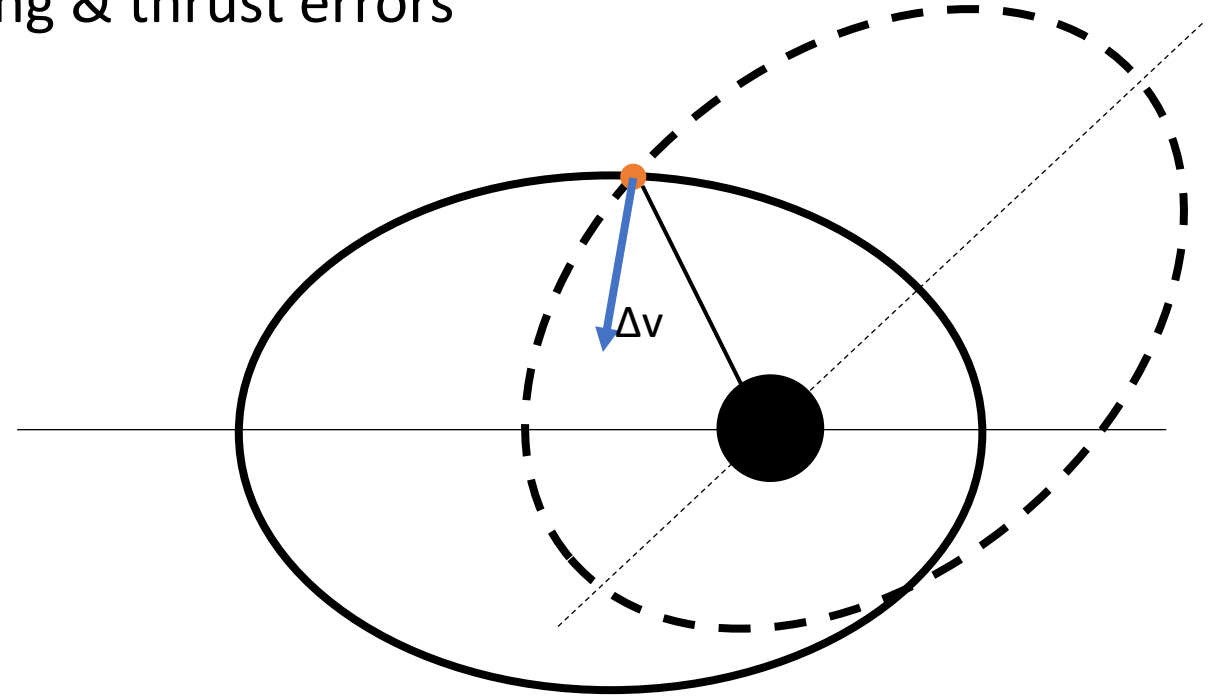


- Special case: no apse line rotation
 - Simpler expression for $\theta_1 = \arccos \left(\frac{h_1^2 - h_2^2}{e_1 h_2^2 - e_2 h_1^2} \right)$
 - Rest remains unchanged
- Coaxial transfer with single impulse
 - Less efficient than Hohmann
 - Requires only one burn
- Useful if constraints prohibit Hohmann
 - Single use boosters
 - Time, exclusion zones, operational requirements



Inverse problem: given a Δv maneuver, what's the new orbit

- Relevant for sensitivity analysis to pointing & thrust errors



Inverse problem: given a Δv maneuver, what's the new orbit

- Relevant for sensitivity analysis to pointing & thrust errors

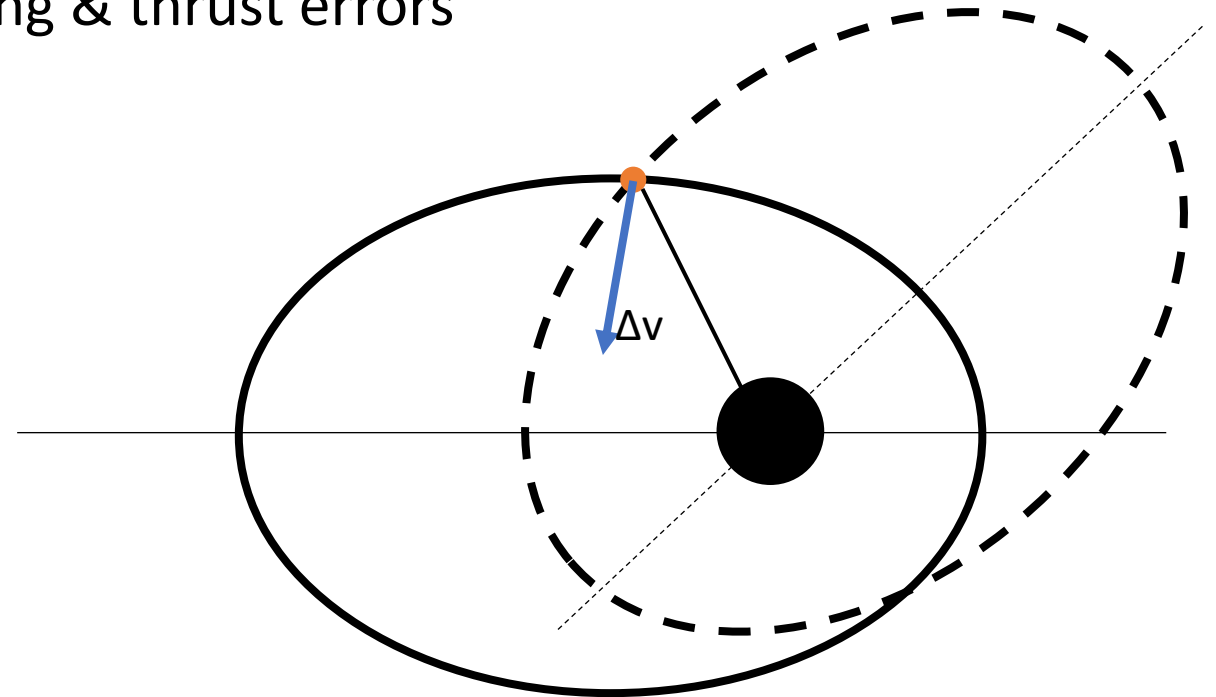
Known: $e_1, h_1, \theta_1, \Delta v_r, \Delta v_\perp$

1. New angular momentum:

$$h_2 = h_1 + r \Delta v_\perp$$

2. New radial velocity yields:

$$v_{2,r} = v_{1,r} + \Delta v_r = \frac{\mu}{h_2} \boxed{e_2 \sin(\theta_2)}$$



Orbit from arbitrary impulse

Inverse problem: given a Δv maneuver, what's the new orbit

- Relevant for sensitivity analysis to pointing & thrust errors

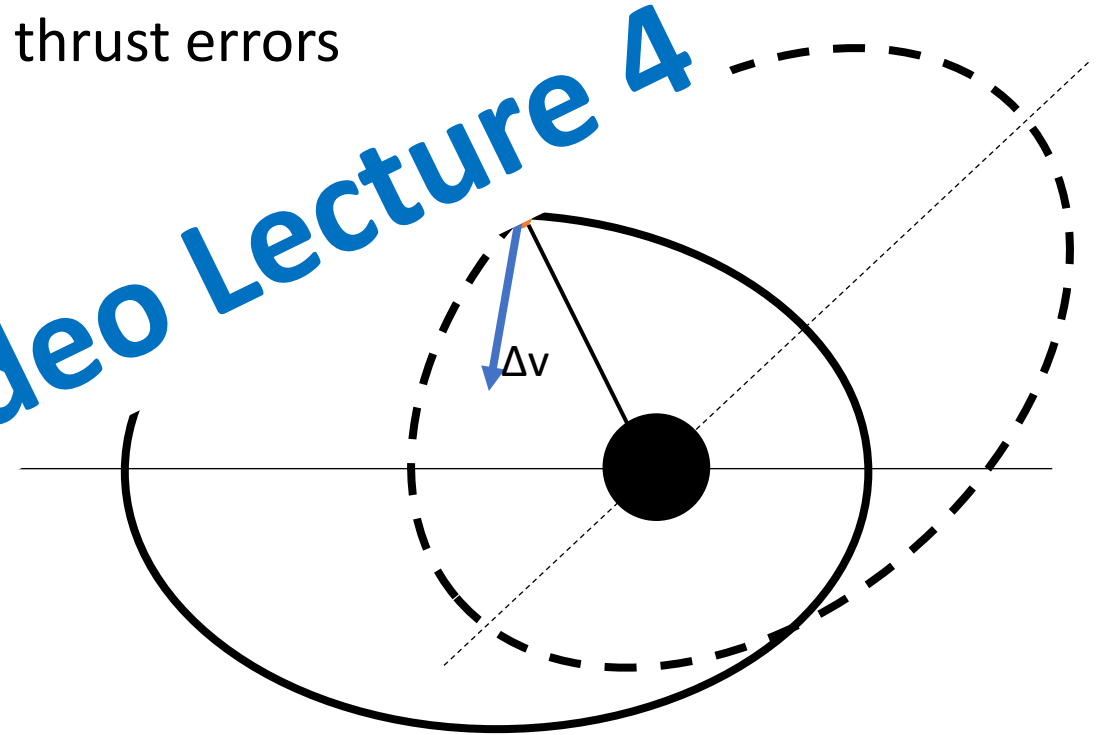
Known: $e_1, h_1, \theta_1, \Delta v_r, \Delta v_\perp$

1. New angular momentum:

$$h_2 = h_1 + r \Delta v_\perp$$

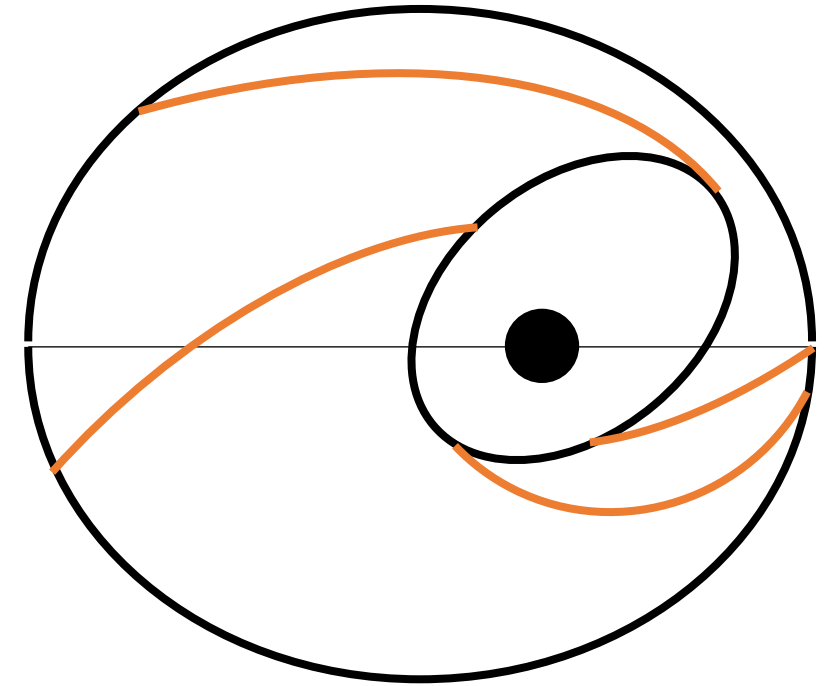
2. New radial velocity yields:

$$v_{2,r} = v_{1,r} + \Delta v_r = \frac{\mu}{h_2} \boxed{e_2 \sin(\theta_2)}$$



see Chapter 1 Video Lecture 4

- Hohmann and Non-Hohmann basic building blocks
- maneuvers can be combined:
multi-maneuver transfers
 - Example: Non-Hohmann transfer orbit (2 impulse)
 - Lambert transfers
- Not obvious which is optimal
- Numerical optimization:
 - Parametrize transfers
 - Find optimal values through optimization



Exact maneuvers depends on problem.
Think before applying equations. Practice!

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